Geometry: Parallelogram Properties

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DIRECTIONS *Use the four properties of parallelograms to solve each problem. Show your equation setup and all work.*

1 Opposite sides are equal. Find x:

$$\overline{AB} = x + 7, \quad \overline{CD} = 2x + 2$$

Answer: ______________________

2 In parallelogram PQRS, PQ = RS. Find x:

$$PQ = 4x + 2, \quad RS = 2x + 6$$

Answer: ______________________

3 Find the perimeter of the parallelogram:

$$AB = 14, \quad BC = 3$$

Answer: ______________________

4 Opposite angles of a parallelogram are equal. Find x:

$$\angle A = (2x + 8)^\circ, \quad \angle C = (x + 26)^\circ$$

Answer: ______________________

5 Find the opposite angle in the parallelogram:

$$\angle A = 115^\circ$$

Answer: ______________________

6 Opposite angles of a parallelogram are equal. Find x:

$$\angle A = (2x + 14)^\circ, \quad \angle C = (x + 32)^\circ$$

Answer: ______________________

7 Find the consecutive angle (supplementary) in the parallelogram:

$$\angle A = 75^\circ$$

Answer: ______________________

8 Consecutive angles are supplementary. Find x:

$$\angle A = (5x + 6)^\circ, \quad \angle B = (2x + 132)^\circ$$

Answer: ______________________

9 Find x and angle Q (consecutive angles are supplementary):

$$\angle P = (3x + 44)^\circ, \quad \angle Q = ?$$

Answer: ______________________

10 Diagonals bisect each other. Find x:

$$\overline{AO} = 4x + 3, \quad \overline{OC} = x + 24$$

Answer: ______________________

11 Diagonals bisect each other. Find x:

$$\overline{BO} = 3x + 7, \quad \overline{OD} = x + 19$$

Answer: ______________________

12 The diagonals of a parallelogram bisect each other. Find AC:

$$\overline{AO} = 12$$

Answer: ______________________

Answer Key & Solutions

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TEACHER NOTES

1) Opposite sides equal: AB=CD, BC=AD. 2) Opposite angles equal: angle A = angle C, angle B = angle D. 3) Consecutive angles supplementary: angle A + angle B = 180. 4) Diagonals bisect each other: AO = OC, BO = OD.

1 Opposite sides are equal. Find x:

$\overline{AB} = x + 7, \quad \overline{CD} = 2x + 2$

= $x = 5$

Set x+7 = 2x+2, solve for x = 5.

2 In parallelogram PQRS, PQ = RS. Find x:

$PQ = 4x + 2, \quad RS = 2x + 6$

= $x = 2$

Opposite sides are equal: 4x+2 = 2x+6, x = 2.

3 Find the perimeter of the parallelogram:

$AB = 14, \quad BC = 3$

= $P = 34$

P = 2(AB+BC) = 2(14+3) = 2(17) = 34.

4 Opposite angles of a parallelogram are equal. Find x:

$\angle A = (2x + 8)^\circ, \quad \angle C = (x + 26)^\circ$

= $x = 18$

Set 2x+8 = x+26, solve: x = 18.

5 Find the opposite angle in the parallelogram:

$\angle A = 115^\circ$

= $\angle C = 115^\circ$

Opposite angles are equal: angle C = angle A = 115 degrees.

6 Opposite angles of a parallelogram are equal. Find x:

$\angle A = (2x + 14)^\circ, \quad \angle C = (x + 32)^\circ$

= $x = 18$

Set 2x+14 = x+32, solve: x = 18.

7 Find the consecutive angle (supplementary) in the parallelogram:

$\angle A = 75^\circ$

= $\angle B = 105^\circ$

Consecutive angles sum to 180: angle B = 180 - 75 = 105 degrees.

8 Consecutive angles are supplementary. Find x:

$\angle A = (5x + 6)^\circ, \quad \angle B = (2x + 132)^\circ$

= $x = 6$

Set (5x+6)+(2x+132) = 180, solve: x = 6.

9 Find x and angle Q (consecutive angles are supplementary):

$\angle P = (3x + 44)^\circ, \quad \angle Q = ?$

= $x = 7, \quad \angle Q = 115^\circ$

3x+44 = 65 → x = 7. Q = 180 - 65 = 115 degrees.

10 Diagonals bisect each other. Find x:

$\overline{AO} = 4x + 3, \quad \overline{OC} = x + 24$

= $x = 7$

AO = OC (diagonals bisect): 4x+3 = x+24, x = 7.

11 Diagonals bisect each other. Find x:

$\overline{BO} = 3x + 7, \quad \overline{OD} = x + 19$

= $x = 6$

BO = OD: 3x+7 = x+19, x = 6.

12 The diagonals of a parallelogram bisect each other. Find AC:

$\overline{AO} = 12$

= $\overline{AC} = 24$

AC = 2 x AO = 2 x 12 = 24 (O is the midpoint).